Introduction

In the last chapter we studied that anything around us is matter and it is made up of molecules. The molecules are combination of atoms of different elements or the same element. Table, chair, bag, book, chalk and blackboard, in short everything you see around are made up of atoms. Atoms are the smallest particles. They cannot be seen even through a microscope. In this lesson, we are going to study about atomic theories, sub-atomic particles, atomic number and mass number and valency.

4.1 Atomic Theories

An atom is thousand times smaller than the thickest human hair. It has an average diameter of 0.000000001 m or 1×10^{-9} m. To understand the size of an atom, now let us find what is the size of known things like pencil, red blood cell, virus and dust particle.



Pencil (1×10^{-2} m)



Red Blood Cell (1×10^{-4} m)



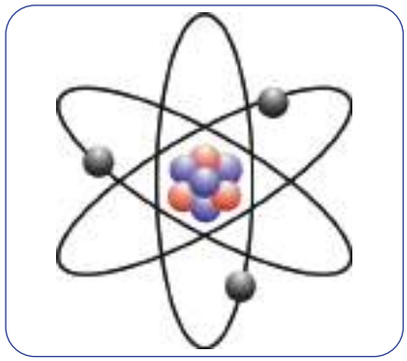
Virus (1×10^{-6} m)



Dust Particle (1×10^{-7} m)

Now you could imagine how small an atom would be.

Many scientists have studied the structure of the atom and advanced their theories about it. The theories proposed by Dalton, Thomson and Rutherford are given below.



Atom (1×10^{-10} m)

4.1.1 Dalton's atomic theory



John Dalton

John Dalton proposed an atomic theory in the year 1808. He proposed that matter consists of very small particles which he named atoms. An atom is the smallest indivisible particle. It is spherical in shape. His theory does not propose anything about the positive and negative charges of an atom. Hence, it was not able to explain many of the properties of substances.



Nanometer is the smallest unit used to measure small lengths. One nanometer is equal to 1×10^{-9} m.

4.1.2 Thomson's theory



J.J. Thomson

In 1897 J.J Thomoson proposed a different theory. He compared an atom to a watermelon. His theory proposed that an atom has positively charged part like the red part of the watermelon and in it are embedded, like the seeds, negatively charged particles

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